

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 12



By- Vishal Sir



Recap of Previous Lecture



- ✓ **Topic** Normal forms
- ✓ **Topic** First normal form (1NF)
- ✓ **Topic** Redundancy in relation because of FD



Topics to be Covered



✓
Topic

Redundancy in relation because of FD

Topic

Second normal form (2NF)

Topic

Third normal form (3NF)

Topic

Boyce codd normal form (BCNF)

✓
Topic

Decomposition of relation



Topic : Normalization



Normalization is the process of reducing/eliminating the redundancy present in the relation.



Topic : Normal forms



There are various normal forms

1 NF, 2NF, 3NF, BCNF
(Boyce Codd Normal Form)

4 NF

- Upto BCNF we will try to eliminate the redundancy present in the relation because of functional dependency.
- If the relation is in BCNF, then there will be 0% redundancy in that relation because of functional dependency but redundancy may be present in that relation because of Multi-valued dependency.

- 4th normal form is related to Multi-valued dependency.
- In 4NF we will try to eliminate the redundancy present in the relation because of Multi-valued dependency.



Topic : Normal forms





Topic : First normal form (1NF)

A database is in 1NF only if it does not contain any multi-valued attributes.

it is multi-valued attribute

Sid	Proj-no.
S ₁	{ P ₁ , P ₂ }
S ₂	{ P ₂ , P ₃ }
S ₃	P ₁

Bring the DB into 1NF
{ i.e., Convert multi-valued attribute into single valued attribute }

Multi value attribute is present in database, so it is not in 1NF

Sid	Proj-no.
S ₁	P ₁
S ₁	P ₂
S ₂	P ₂
S ₂	P ₃
S ₃	P ₁

Both are single valued attributes

No multi-valued attribute is present in the database, so it is in 1NF

* By default normal form a relation is "1NF"
i.e, Every relation is at least in 1NF

1NF does not eliminate redundancy,
it only converts multi-valued attributes
into single valued attributes.



Topic : Redundancy in relation because of FD

1st Rule : A non-trivial functional dependency $X \rightarrow Y$ does not cause any redundancy in the relation if "X" is a Super Key of that relation.



Topic : Redundancy in relation because of FD

2nd Rule : A non-trivial functional dependency $X \rightarrow Y$ will cause redundancy in the relation if "X" is not a super key of that relation

$A \rightarrow BC$

|||

$A \rightarrow B$

$A \rightarrow C$

A is super key
there is

no redundancy

w.r.t $A \rightarrow B$

$A \rightarrow C$

A	B	C
1	1	4
2	2	7
3	1	4
4	1	4
5	2	7
6	3	7

$B \rightarrow C$

and 'B' is not a s.k of relation.

∴ There is redundancy
in the relation

w.r.t. $B \rightarrow C$

Note:- In non-trivial functional dependency $X \rightarrow Y$, if "X" is not a super key of that relation, then "Y" can never be the superkey of that relation.

∴ A non-trivial functional dependency $X \rightarrow Y$ in which none of X or Y is a Super Key will always cause redundancy in the relation.

→ If neither X is a Super key nor Y is a Super key, then possible choices for X and Y are,

- (i) X & Y may be any Collection of non-prime attributes
- (ii) X & Y may be proper subsets of C.K.
- (iii) X & Y may be Combination of Non-prime attribute & Proper subset of Candidate Key

X $\xrightarrow{\hspace{1cm}}$ Y

P.S.C.K



N.P.A.

P.S.C.K



P.S. of another C.K.

~~X - P.S.C.K~~

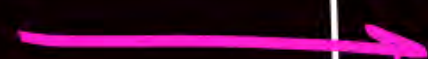


$(N.P.A + P.S.C.K)$

$\equiv$

$P.S.C.K \rightarrow N.P.A.$
 $P.S.C.K \rightarrow P.S. \text{ another } C.K.$

N.P.A.



N.P.A.

~~X~~

N.P.A



P.S.C.K.

Such FD is not possible

~~X - N.P.A~~



$(N.P.A + P.S.C.K)$

$\equiv$

$N.P.A \rightarrow N.P.A.$
 $N.P.A \rightarrow P.S.C.K$

$(N.P.A + P.S.C.K)$



N.P.A.

$(N.P.A + P.S.C.K)$



P.S. another C.K.

~~X - $(N.P.A + P.S.C.K)$~~



$(N.P.A + P.S. \text{ of another } C.K.)$

* Possible types of non-trivial functional dependencies that may cause redundancy in the relation

X $\longrightarrow$ Y

- | | | | |
|---------|----------------------------------|-------------------|------------------------------|
| Case ①: | (P.S.C.K) Proper Subset of a C.K | $\longrightarrow$ | Non Prime Attribute (N.P.A.) |
| Case ②: | (N.P.A.) Non prime Attribute | $\longrightarrow$ | Non prime Attribute (N.P.A.) |
| Case ③: | (N.P.A + P.S.C.K.) | $\longrightarrow$ | N.P.A. |
| Case ④: | P.S. of one C.K | $\longrightarrow$ | P.S. of another C.K |
| Case ⑤: | (N.P.A + P.S. of one C.K) | $\longrightarrow$ | P.S. of another C.K. |

Case Normal Form	Case① FD	Case② FD	Case③ FD	Case④ FD	Case⑤ FD
1NF	✓	✓	✓	✓	✓
2NF	✗ Not allowed	✓	✓	✓	✓
3NF	✗ Not allowed	✗ Not allowed	✗ Not allowed	✓	✓
BCNF	✗	✗	✗	✗	✗

} All types of FDs are allowed in 1NF

} FDs w/ Case①, Case② & Case③ are not allowed in 3NF

FDs of type Case① are not allowed in 2NF

All 5 types of FDs are not allowed in BCNF



Topic : Second normal form (2NF)

A relation R is in 2NF only if,

① R is in 1NF

and ② Relation R must not contain any partial functional dependency

Partial Functional dependency:

- * If proper subset of a candidate key can determine a non-prime attribute, then partial functional dependency exists in the relation.

eg Let Candidate key of Relation R is (AB), and let 'C' is a non-prime attribute of relation R.

If functional dependency $\underbrace{B}_{\text{P.S.C.K.}} \rightarrow \underbrace{C}_{\text{N.P.A.}}$ exists in R, then "AB $\rightarrow$ C" is a partial dependency.

Note:- If functional dependency of type " $P.S.CK \rightarrow N.P.A$ "
{i.e. FD w.r.t. Case ①} exist in relation "R" then
Partial functional dependency will also exist
in relation R.

∴ For relation "R" to be in 2NF it must not
contain a functional dependency same as Case ①.



Topic : Third normal form (3NF)

A relation R is in 3NF only if
Every non-trivial functional dependency
with

$X \rightarrow Y$ is

(i) X as a S.K. of the relation

(or)

(ii) Y as a prime attribute of relation

i.e. The relation must not contain any functional dependency same as Case① or Case② or Case③



Topic : Third normal form (3NF)

A relation R is in 3NF only if
relation R is in 2NF and
it must not contain any transitive functional dependency

C.K / S.K $\longrightarrow$ Non-prime Attribute

In this case
non-prime attribute
is directly dependent
on C.K, or
No transitive FD

Super Key₁ $\longrightarrow$ Super Key₂

Super Key₂ $\longrightarrow$ Non-prime Attribute

In this case also
the non-prime attribute
is directly dependent
on a Super Key₂

A functional dependency
is call transitive functional
dependency if non-prime
attribute is transitively
dependent on ^{Candidate Key} (or) ^{Super Key}
i.e., Not directly dependent
on C.K.

$S.K. \longrightarrow N.P.A_1$

$N.P.A_1 \longrightarrow N.P.A_2$

In this case $N.P.A_1$ is directly dependent on Super Key.

But,

$N.P.A_2$ is not directly dependent on S.K. {i.e. transitively dependent on S.K.}

∴ FDs of type

$N.P.A_1 \longrightarrow N.P.A_2$ (Case ②) are not allowed in 3NF

$(P.S.C.K + N.P.A_1) \longrightarrow N.P.A_2$

In this case also $N.P.A_2$ is not directly dependent on Super Key.

∴ FDs of type

$(P.S.C.K + N.P.A_1) \longrightarrow N.P.A_2$ (Case ③) are also not allowed in 3NF



Topic : Boyce codd normal form (BCNF)

For a relation to be in BCNF, there must be no redundancy in the relation because of functional dependency. { $\therefore$ All 5 types of FDs are not allowed } in a BCNF relation

- * For a relation R to be in BCNF, Every non-trivial functional dependency $X \rightarrow Y$ must have " X " as a Super key of relation R .

#e.g.

Given $R(ABCDEF)$ and

$$F = \{AB \rightarrow CD, D \rightarrow A, C \rightarrow E, B \rightarrow F\}$$

Find the normal form of the relation.

Normal form of a relation is least of the highest normal form satisfied by each FD of the relation. $\therefore$ Normal form of given relation is "1NF".

$\underbrace{AB}_{\text{S.K}} \longrightarrow CD$ { BCNF }

{ Such FDs allowed }
{ upto BCNF }

$D \xrightarrow{\quad} A$
 P.S. of one C.K. $\xrightarrow{\text{3NF}}$ P.S. of another C.K. (Case 4)

{ it is allowed upto 3NF }
 { But not allowed in BCNF }

$C \xrightarrow{\text{N.P.A.}} E$
 $\{2NF\}$ NPA (Case ②)

{ Allowed upto 2NF,
 But not allowed in 3NF }

$B \xrightarrow{\text{P.S. CK}} F$
 $\{1NF\}$ NPA (Case ①)

$\boxed{AB}$ is a C.K. $\left. \begin{array}{l} \text{These are} \\ \text{only two} \\ \text{C.K.s} \end{array} \right\}$
 $\downarrow$
 $\boxed{DB}$ is a C.K. $\left. \begin{array}{l} \text{C.K.s} \end{array} \right\}$

$$P.A. = \{A, B, D\}$$
$$N.P.A. = \{C, E, F\}$$

#e.g. Given $R(ABCDE)$ and $F=\{AB \rightarrow C, C \rightarrow D, B \rightarrow E\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

CK: AB

P.A: $\{A, B\}$

N.P.A: $\{C, D, E\}$

AB $\rightarrow$ C {BCNF}
S.K

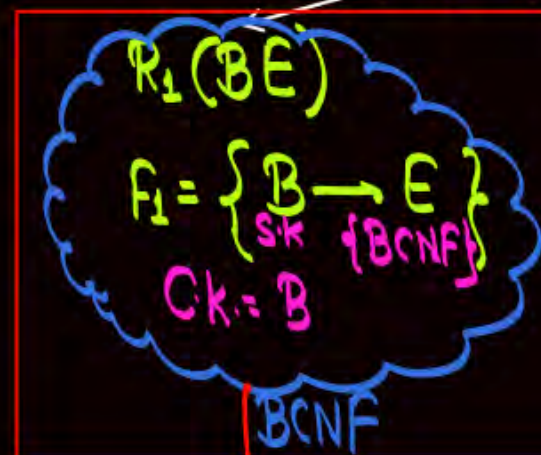
C $\rightarrow$ D {2NF}
N.P.A Case 2 N.P.A

B $\rightarrow$ E {1NF}
P.S.CK Case 1 N.P.A.

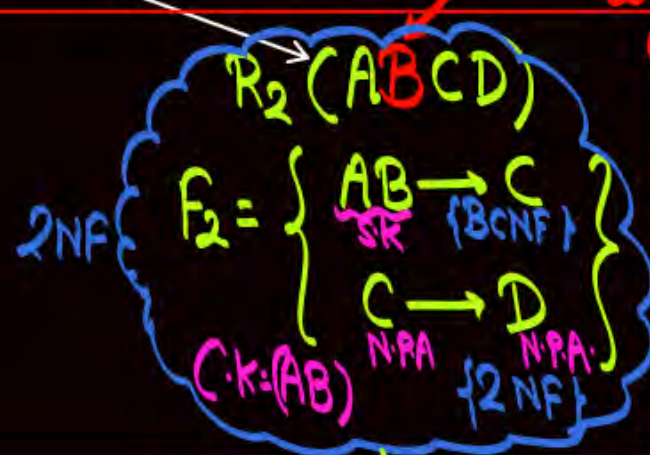
Normal form of Relation is "1NF"

$R(A, B, C, D, E)$ is not in 2NF
 because of FD " $B \rightarrow E$ "
 ∴ Decompose w.r.t. $B \rightarrow E$

$$(BE)^+ = \{B, E\}$$



already in BCNF



"B" is added
 to ensure
 lossless join
 decomposition

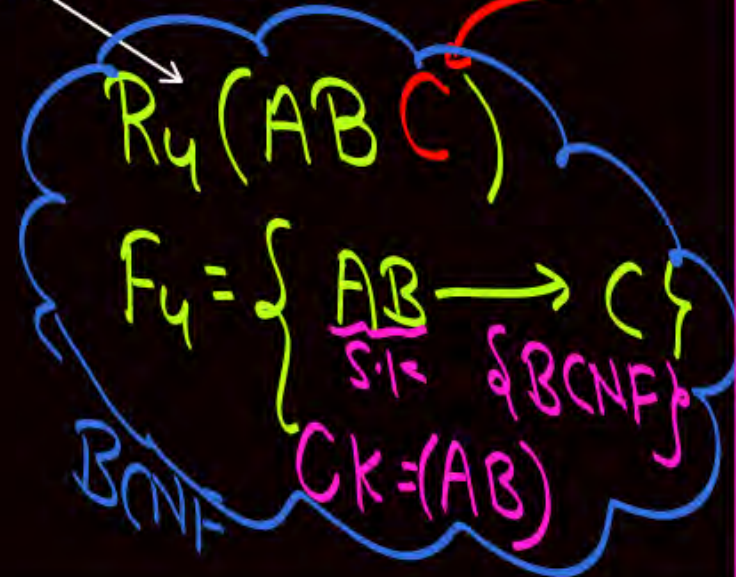
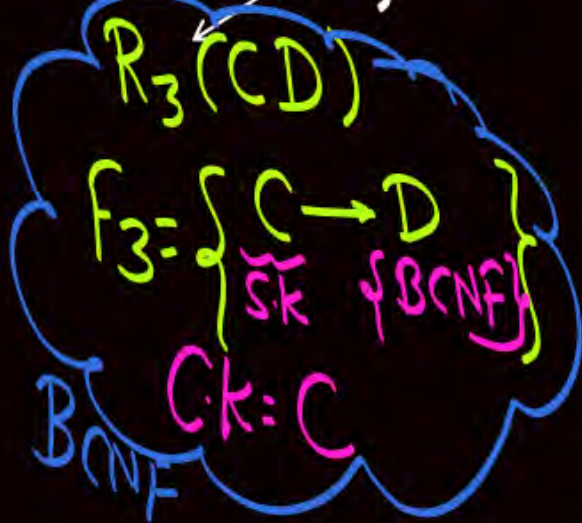
Overall
 normal form
 is = 2NF

+
 Lossless Join
 +
 Dep. preserving

$R_2(ABCD)$ is not in 3NF

Because of $C \rightarrow D$.

∴ Decompose w.r.t. $C \rightarrow D$



added to ensure
 lossless join
 3NF as well as BCNF
 +
 Lossless join decomposition
 +
 Dep. preserving decomposition

H.W.

#e.g.



Given $R(ABCDEF)$ and $F = \{A \rightarrow BCDEF, BC \rightarrow ADEF, D \rightarrow E, E \rightarrow F\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.
#e.g.



Given $R(ABCD)$ and $F=\{AB \rightarrow C, BC \rightarrow D\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.



#e.g.

Given $R(ABCDEFGH IJ)$ and $F=\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.



#e.g.

Given $R(ABDLPT)$ and $F=\{B \rightarrow PT, T \rightarrow L, A \rightarrow D\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.



#e.g.

Given $R(ABCDE)$ and $F:\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

H.W.

#e.g.

Given $R(ABCD)$ and $F=\{AB \rightarrow CD, D \rightarrow A\}$



Find the normal form of the relation, and if relation is not already in BCNF then decompose the relation up to BCNF.

THANK - YOU